

MD SOAD SOLAIMAN

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EDUCATION

- **Master of Science in Data Science** *Jan 2025 - Present (Expected Nov 2026)*
Adelaide University, Adelaide, Australia, GPA: 6.2 / 7.0
- **Bachelor of Science in Industrial & Production Engineering** *Jan 2020 - Mar 2024*
Military Institute of Science and Technology (MIST), BUP, Bangladesh, CGPA: 3.69 / 4.00

RESEARCH & TECHNICAL PROJECTS

MIT-BIH Arrhythmia Heartbeat Classification *Coursework Research Project, 2026*

- Built a record level pipeline classifying four AAMI heartbeat classes from the MIT-BIH Arrhythmia Database (physionet.org), comparing Logistic Regression, Random Forest, and CNN models.
- Evaluated models using per-class precision/recall/F1, precision-recall AUC, and McNemar's test for statistical comparison of minority-class recall, relating model complexity to minority-class performance.

<https://github.com/mdsoadsolaiman/MIT-BIH-Heartbeat-Classification>

Machine Learning & Deep Learning Portfolio *Projects 2026*

- Blood-Cell ResNet Classification: Built a ResNet-based CNN pipeline for eight-class image classification, including model architecture design, training, and checkpoint-based evaluation.
- Diabetes MLP Classification: Developed multilayer perceptron (MLP) models for binary tabular classification, applying feature preprocessing and ROC-AUC-based model evaluation.
- MNIST Classification and Clustering: Applied Logistic Regression and RBF-kernel SVM for supervised digit classification with cross-validation, and PCA with K-Means for unsupervised clustering.
- Butterfly CNN Classification: Designed and trained a compact convolutional neural network (CNN) for five-class image classification, including data preprocessing and held-out test evaluation.

<https://github.com/mdsoadsolaiman/Machine-Learning-and-Deep-Learning-Projects>

Trustworthy Foundation Models for Time-Series Forecasting *Research, 2026*

- Developed a leakage-controlled, reproducible forecasting framework comparing Chronos, TimesFM, classical models (ARIMA, SARIMA, Holt winters, exponential smoothing), LSTM, Prophet and naive models across Bitcoin and electricity datasets.
- Evaluated models using accuracy, statistical significance, robustness, temporal stability, uncertainty calibration, and multidimensional trustworthiness analysis.

<https://github.com/mdsoadsolaiman/Time-Series-Foundation-Models>

Advanced Analytics Portfolio *Projects 2026*

- Business Analytics: Applied KPI analysis, financial comparison, and investment evaluation techniques across venue performance, player compensation, and stadium profitability datasets.
- Public Data Analytics: Built an interactive R/Quarto/Shiny application, applying data cleaning, geographic comparison, and affordability analysis to public rental market data.
- Statistical Research Analytics: Conducted a staged statistical workflow in R/Quarto — exploratory data analysis, hypothesis testing, regression modelling, and model diagnostics.

<https://github.com/mdsoadsolaiman/Advanced-Analytics>

TECHNICAL SKILLS

- **Programming and libraries:** Python (PyTorch, TensorFlow, Keras, scikit-learn, pandas, NumPy, SciPy, OpenCV); R (tidyverse, tidymodels, caret, and others)
- **Machine learning:** Regression (linear, ridge, lasso, logistic), decision trees, random forests, support vector machines, K-means clustering, ensemble methods (bagging, boosting, AdaBoost, stacking)
- **Deep learning and representation learning:** MLPs, CNNs, ResNets, RNNs, LSTMs; transfer learning, fine-tuning, self-supervised and contrastive learning
- **Time-series forecasting and foundation models:** ARIMA, SARIMA, exponential smoothing, Prophet, and foundation models including Chronos and TimesFM; zero-shot and multi-step forecasting, cross-domain generalization, uncertainty calibration
- **Data preprocessing and feature engineering:** Cleaning, missing-value handling, normalization, feature selection and extraction, PCA, SVD, t-SNE, and image augmentation
- **Model training and optimization:** Hyperparameter tuning, cross-validation, gradient descent, Adam/AdamW, early stopping, dropout, class weighting, learning-rate scheduling
- **Model evaluation and statistical inference:** Precision, recall, F1, ROC-AUC, calibration, ablation studies, leakage prevention, bootstrap confidence intervals, hypothesis testing, ANOVA
- **Data visualization and reporting:** Tableau, Shiny, Plotly, Quarto, R Markdown, Jira, and Microsoft Office Suite
- **Research and development tools:** Git/GitHub, Jupyter, HTML, CSS, and statistical model evaluation

PUBLICATIONS

- Solaiman, M. S., Sakib, M. S. I., Rahman, M. A., & Rashid, A. B. (2024). Effect of cutting parameters on performance of CNC hot wire styrofoam cutting machine. December 2024.
DOI: 10.46254/BA07.20240050
- Mahin, A. M. S., Solaiman, M. S., & Shahriar, B. M. (2024). Optimization of inbound logistics of automobile industry using FlexSim. December 2024.
DOI: 10.46254/BA07.20240083

PROFESSIONAL EXPERIENCE

Assistant Engineer, Business Operations & Development Feb 2024 - Dec 2024

Anwar Group of Industries, Bangladesh

- Supported business operations and process-development initiatives across manufacturing units, applying industrial engineering methods to operational problems.

LEADERSHIP & UNIVERSITY SERVICE

Peer Connect Guide Jan 2026 – Present

Adelaide University

- Support and mentor incoming postgraduate students during their transition to university life and study at Adelaide.

President, MIST Career Club Jan 2023 - Jun 2023

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

- Led club operations, coordinated career-development initiatives, and managed collaboration among students, faculty, and external stakeholders.